

YogaGuruGPT: A Specialized LLM-Based Yoga Education Assistant Using RAG

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GitHub Link - <https://github.com/akshat-mishra-ds/YogaGuruGPT>

Abstract

The rapid growth of Artificial Intelligence and Large Language Models (LLMs) has transformed the way users access and interact with information. However, general-purpose language models often lack domain-specific knowledge and may provide inaccurate or incomplete responses when dealing with specialized subjects. Yoga education is one such domain where knowledge is distributed across multiple books, scriptures, and educational resources, making it difficult for learners to access relevant information efficiently.

This project presents **YogaGuruGPT**, a domain-specific conversational assistant developed for the Education and Training industry. The system utilizes a Retrieval-Augmented Generation (RAG) architecture that combines semantic document retrieval with a pre-trained language model to provide context-aware answers to yoga-related questions. A collection of authoritative yoga texts, including Yoga Anatomy, Bhagavad Gita, Hatha Yoga Pradipika, Prana and Pranayama, Inside the Yoga Sutras, and Shiva Samhita, was used as the knowledge base.

The documents were processed, chunked into smaller sections, converted into vector embeddings using Sentence Transformers, and stored within a FAISS vector database. User queries are matched against the vector database to retrieve relevant content, which is then supplied to the TinyLlama language model for answer generation. A Gradio-based user interface was developed to enable interactive conversations.

The project successfully demonstrates the practical application of Retrieval-Augmented Generation for educational assistance and domain-specific knowledge retrieval while avoiding the computational expense of model fine-tuning.

1. Introduction

Large Language Models have become increasingly capable of understanding and generating natural language. Despite their impressive performance, these models often suffer from limitations when responding to highly specialized or domain-specific

questions. General-purpose models may generate hallucinated responses or fail to provide information grounded in authoritative sources.

Yoga education encompasses a broad range of topics including anatomy, breathing techniques, meditation, classical philosophy, spiritual teachings, and physical postures. Students and practitioners often need to consult multiple books to obtain comprehensive answers to their questions. This process can be time-consuming and inconvenient.

To address this challenge, YogaGuruGPT was developed as a specialized conversational assistant capable of retrieving information from authentic yoga literature and generating user-friendly responses. The project demonstrates how Retrieval-Augmented Generation can improve answer accuracy by grounding responses in trusted source documents.

2. Problem Statement

Yoga learners often face difficulty in locating relevant information from multiple books and educational resources. Traditional search methods require users to manually browse large volumes of content, making learning inefficient.

The objective of this project is to develop a domain-specific chatbot capable of answering yoga-related questions by retrieving and utilizing information from authoritative yoga literature. The chatbot should provide accurate, context-aware, and educational responses while remaining computationally efficient and easy to deploy.

3. Objectives

The primary objectives of this project are:

- Develop a domain-specific AI chatbot for yoga education.
 - Utilize authoritative yoga literature as the knowledge base.
 - Implement Retrieval-Augmented Generation (RAG) architecture.
 - Generate context-aware responses using a pre-trained language model.
 - Create an interactive chatbot interface for end users.
 - Demonstrate practical application of LLMs in the Education and Training industry.
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4. Knowledge Base Description

The following yoga resources were used:

Yoga Anatomy

Provides anatomical explanations of yoga postures, breathing mechanisms, and body movements.

Bhagavad Gita

Contains philosophical teachings, Karma Yoga, Bhakti Yoga, and spiritual wisdom.

Hatha Yoga Pradipika

Traditional text covering asanas, pranayama, mudras, and yogic practices.

Prana and Pranayama

Explains breathing techniques, pranic energy systems, and respiratory practices.

Inside the Yoga Sutras

Provides detailed explanations of Patanjali's Yoga Sutras and yogic psychology.

Shiva Samhita

Classical yoga scripture discussing meditation, kundalini, and spiritual practices.

These documents collectively form the knowledge foundation of YogaGuruGPT.

5. Methodology

The project follows a Retrieval-Augmented Generation workflow.

Step 1: Document Loading

All PDF documents were loaded using LangChain's PyPDFLoader. Each page was converted into a document object for processing.

Step 2: Text Chunking

Documents were divided into smaller overlapping chunks.

Parameters used:

- Chunk Size: 1000 characters
- Chunk Overlap: 200 characters

Chunking improves retrieval efficiency and preserves contextual information.

Step 3: Embedding Generation

Text chunks were transformed into dense vector representations using:

all-MiniLM-L6-v2

This embedding model captures semantic meaning and enables similarity-based retrieval.

Step 4: Vector Database Creation

FAISS (Facebook AI Similarity Search) was used to store embeddings and perform efficient semantic search.

Step 5: Retrieval Mechanism

For each user query:

1. Query embedding is generated.
2. Similar document chunks are retrieved.
3. Most relevant chunks are selected using Maximum Marginal Relevance (MMR).

Step 6: Language Model Integration

The retrieved context is supplied to:

TinyLlama-1.1B-Chat-v1.0

The model generates responses grounded in the retrieved knowledge.

Step 7: User Interface

A Gradio interface was developed to provide a simple and interactive conversational experience.

6. System Architecture

User Query ↓ Embedding Generation ↓ FAISS Vector Database ↓ Relevant Document Retrieval ↓ Prompt Construction ↓ TinyLlama Language Model ↓ Generated Response ↓ User Interface

This architecture ensures that answers are generated using retrieved knowledge rather than relying solely on the language model's internal parameters.

7. Technologies Used

Technology	Purpose
Python	Development Language
Google Colab	Development Environment
LangChain	RAG Framework
FAISS	Vector Database
Sentence Transformers	Embedding Generation
TinyLlama	Language Model
Gradio	User Interface
PyPDFLoader	PDF Processing

8. Results and Evaluation

The chatbot was tested using multiple yoga-related queries.

Sample Query 1

Question: What is Pranayama?

Result: The chatbot successfully retrieved relevant information from the Prana and Pranayama text and generated an educational response.

Sample Query 2

Question: Explain Karma Yoga.

Result: The chatbot retrieved content related to the Bhagavad Gita and generated a context-aware explanation.

Sample Query 3

Question: What is Kapalbhati?

Result: The chatbot correctly identified relevant breathing techniques from the knowledge base.

Evaluation Summary

Metric	Observation
Document Retrieval	Successful
Semantic Search	Accurate
Response Generation	Context-Aware

Metric	Observation
Domain Relevance	High
User Interaction	Smooth

The evaluation demonstrates that the chatbot effectively retrieves and utilizes yoga-related knowledge to answer user queries.

Screenshots:


YogaGuruGPT

A Specialized LLM-Based Yoga Education Assistant

Ask questions about: • Yoga Anatomy • Pranayama • Yoga Sutras • Bhagavad Gita • Meditation • Hatha Yoga

Ask a Yoga Question

What is Pranayama ?

Clear Submit

YogaGuruGPT Answer

Pranayama is a practice that involves controlled breathing and physical postures to promote relaxation, balance, and overall well-being. It is a form of yoga that originated in India and has been practiced for thousands of years. Pranayama involves deep breathing exercises, such as inhaling for a count of four, holding for a count of four, and exhaling for a count of four, while focusing on the breath and the sensation of the breath moving in and out of the body. The practice aims to calm the mind and body, reduce stress and anxiety, and promote overall well-being.


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Ask a Yoga Question

Explain Karma Yoga

Clear Submit

YogaGuruGPT Answer

Karma Yoga is a practice that involves performing actions with the intention of creating positive results in the present and future. It is a way of living that recognizes that our actions have both positive and negative consequences, and that these consequences can be harnessed to create a more fulfilling and peaceful existence.

Karma Yoga is based on the principle that all actions, no matter how small or insignificant, have the potential to create positive or negative outcomes. By performing actions with a sense of purpose and intention, we can create a positive impact on ourselves and the world around us.

One of the key components of Karma Yoga is the practice of selfless service. This involves performing actions without any expectation of reward or recognition, and instead focusing on the joy and fulfillment that comes from helping others. This practice can be done in various ways, such as volunteering at a local charity, helping a friend with a project, or simply offering a kind word or gesture to a stranger.

Another aspect of Karma Yoga is the practice of meditation and mindfulness. By focusing on the present moment and letting go of distractions, we can cultivate a sense of inner peace and clarity. This can be done through various techniques, such as deep breathing, mindfulness meditation, or yoga.

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Ask a Yoga Question

What is Ashtanga Yoga



Clear

Submit

YogaGuruGPT Answer

Ashtanga Yoga is a Hatha Yoga practice that involves a series of 26 postures (or asanas) performed in a specific order for 90 minutes. The practice is designed to balance the body's energy centers (pranic, chi, and prana) and promote physical, mental, and spiritual well-being. The practice is known for its emphasis on breath control, alignment, and mindfulness, and is often practiced in a group setting or with a teacher. Ashtanga Yoga is considered a powerful tool for cultivating inner peace, strength, and flexibility, and has been practiced by many renowned yoga teachers and celebrities.

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Ask a Yoga Question

Tell me about Tree Pose



Clear

Submit

YogaGuruGPT Answer

Tree Pose, also known as Vrksasana, is a standing-down pose that is commonly practiced in yoga. It is a variation of the standing pose, Uttanasana, and is often used to strengthen the legs, hips, and core. The pose is often practiced with arms extended in front of the body, and the head and neck are relaxed. The pose is believed to promote relaxation, balance, and strength in the body. It is a popular pose in yoga classes and is often used to open up the hips and lower back. Tree Pose is a great pose for beginners as it is easy to learn and can be practiced with minimal equipment.

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Ask questions about: • Yoga Anatomy • Pranayama • Yoga Sutras • Bhagavad Gita • Meditation • Hatha Yoga

Ask a Yoga Question

What are the Eight limbs of Yoga?



Clear

Submit

YogaGuruGPT Answer

The Eight Limbs of Yoga are:

1. Yama (abstinence)
2. Niyama (observance)
3. Asana (posture)
4. Pranayama (breath control)
5. Pranayama (breath control)
6. Kriya (purification)
7. Jnana (knowledge)
8. Bhakti (devotion)

The eight limbs are described in the Yoga Sutras of Patanjali, which are considered the foundation of yoga. The final position, gomukhasana, is not mentioned in the final position, but it is included in the final position of the other limbs.

9. Challenges Faced

Several challenges were encountered during development:

- Large PDF documents required significant processing time.
 - Some PDF pages contained formatting inconsistencies.
 - Selecting an appropriate chunk size required experimentation.
 - Smaller language models occasionally generated shorter responses.
 - Balancing retrieval accuracy and response generation quality required prompt engineering.
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10. Limitations

Despite its effectiveness, the system has certain limitations:

- Limited to information contained within uploaded documents.
 - Cannot answer questions outside the yoga domain accurately.
 - Does not maintain long-term conversational memory.
 - Uses a lightweight language model with limited reasoning capabilities compared to larger commercial models.
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11. Future Scope

Several improvements can be implemented:

- Support multilingual conversations.
 - Add voice-based interaction.
 - Incorporate conversational memory.
 - Deploy as a web application.
 - Integrate larger open-source language models.
 - Expand the knowledge base with additional yoga literature and research papers.
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12. Conclusion

YogaGuruGPT successfully demonstrates the use of Retrieval-Augmented Generation for developing a domain-specific educational assistant. By combining authoritative yoga

literature, semantic embeddings, FAISS vector search, and a pre-trained language model, the chatbot provides meaningful and context-aware responses to yoga-related queries.

The project highlights the potential of combining document retrieval and language generation to create practical educational tools. It also demonstrates that domain-specific assistants can be developed effectively without expensive fine-tuning, making the solution scalable, maintainable, and suitable for real-world deployment in the Education and Training industry.